Part I: GRPO Finetuning of `gemma-3-1b-it` on GSM8K

Zixuan Wang (zw499)

Team **chmawa**: sm3035 · bc654 · zw499

Code: github.com/wdk0082/a8-grpo | W&B: a8-grpo

I.1 Reproducing the baseline — the run & patches

(i) The run. Baseline reproduced as run **R0** (`8c2785ut`): the upstream **Tunix / JAX** GRPO pipeline (LoRA on frozen `gemma-3-1b-it`), commit `borisbolliet/tpu-2026@77c5a67`, run with the `config.py` defaults — KL-penalty weight β=0.08, group size K=2 (responses sampled per prompt), the full **3364** GRPO steps (one epoch), ∼**5 h 05 m** wall-clock on `v6e-1`.

Baseline patches (fixes for what did not run as-shipped; full diff in `tpu-2026_our_changes.patch`):

- TFDS's gsm8k builder crashes under protobuf 7.x once JAX imports — we evaluate the *identical* GSM8K test via an HF `datasets` fallback (seed 42);
- raised checkpoint retention (`SAVE_INTERVAL_STEPS` 500→100, `MAX_TO_KEEP` 4→40) so the best-val checkpoint survives selection (persistence, not training).

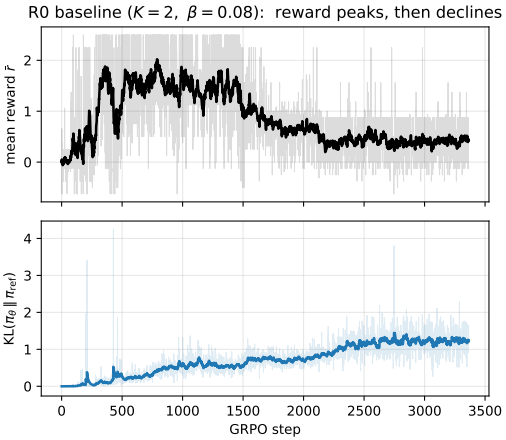
I.1 Reproducing the baseline — accuracy & training curves

(ii) Held-out accuracy. GSM8K *test* (from HF), seed 42, greedy / exact-match, reported as **(**n=64 **/** n=1319**)** — n=64 is the `config.py` default eval, n=1319 the full test:

Model / checkpoint	n=64	n=1319
Base `gemma-3-1b-it`	45.3	47.4
R0 best-val (step 700)	42.2	46.7
R0 final (step 3364)	0.0	6.2

best-val = the saved checkpoint *nearest the peak held-out validation reward* (not accuracy).

(iii) Training curves (R0)



I.2 Understanding the baseline

(a) The three policies (all share the frozen W_0):

- π_θ — LoRA-adapted policy; the only trainable policy ($\Delta W = BA$)
- π_{old} — the rollout policy; with μ=`NUM_ITERATIONS`=1 (one gradient step per rollout batch), $\pi_{\text{old}}=\pi_\theta$ at the update
- π_{ref} — frozen base Gemma

(b) Group size & advantage. K=`NUM_GENERATIONS`=2; Tunix `'grpo'` is the **standard** group-mean, std-normalised advantage over all K (incl. i) — *not* leave-one-out:

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r}, \qquad \hat{g} = \frac{1}{K}\sum_{i=1}^{K} h_i \hat{A}_i, \qquad h_i = \nabla_\theta \log \pi_\theta(a_i \mid q).$$

I.2 Understanding the baseline

(c) Reward = sum of four:

- **Shaping:** `match_format_exactly`, `match_format_approximately` (the `<reasoning>`/`<answer>` template)
- **Correctness:** `check_answer` (gold match, only if the template parses) + `check_numbers` (any number after `<answer>`)

With the correctness reward *alone*, σ_r vanishes on most groups — not because the model is rarely right (base = 47%) but because the reward is **coarse**: at K=2 the two samples usually *tie*, so

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r} = \frac{0}{\sigma_r + 10^{-6}} = 0.$$

The dense **shaping** terms keep $\sigma_r > 0$ and push toward parseable outputs (the precondition for `check_answer` to vary), although they also invite reward-hacking.

I.2 Understanding the baseline

(d) The clipped surrogate. The PPO-style clip lives in Tunix's `GRPOLearner` (`tunix.rl`); the clip range is ϵ=`EPSILON`=0.2 in `config.py`. **Two deviations** from the lecture form:

(i) Token- vs. sequence-level. The lectures clip a **per-response** ratio (one clip per whole sequence); Tunix clips a **per-token** ratio, token-by-token (a token-level objective, as in DAPO):

$$\rho_i = \frac{\pi_\theta(a_i \mid q)}{\pi_{\text{old}}(a_i \mid q)} \qquad \text{vs.} \qquad \rho_{i,t} = \frac{\pi_\theta(a_{i,t} \mid q, a_{i,<t})}{\pi_{\text{old}}(a_{i,t} \mid q, a_{i,<t})}.$$

(ii) The clip is inert in this baseline. With μ=`NUM_ITERATIONS`=1 (one gradient step per rollout batch) $\pi_{\text{old}}=\pi_\theta$, so every $\rho \equiv 1$ and **nothing is ever clipped** — the surrogate collapses to the plain policy gradient and ϵ is irrelevant. Not a *contradiction* (the lecture form also reduces to this at μ=1), but the baseline's "clipped" objective does no clipping until $\mu > 1$.

I.3 Why the baseline collapses: gradient variance

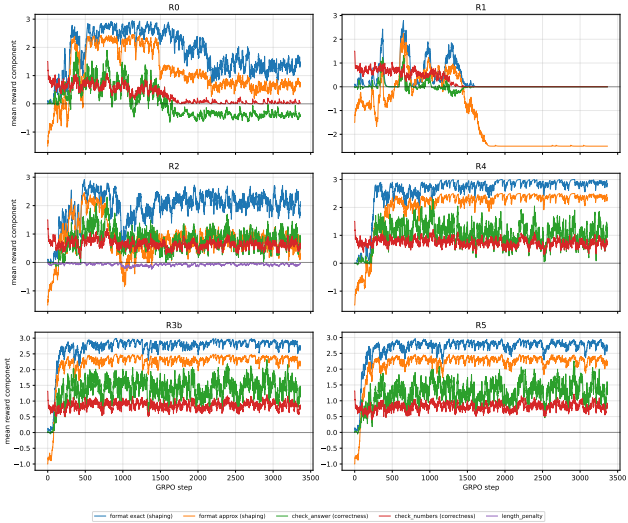
Motivation (grounded in the lecture estimator). The group-mean gradient has variance

$$\mathrm{Var}(\hat{g}) \;=\; \frac{1}{K^2}\sum_i (h_i - \bar{h})(h_i - \bar{h})^\top \;=\; O\!\left(\tfrac{1}{K}\right),$$

- Maximal at the baseline's $K{=}2 \Rightarrow$ noisiest updates
- Noisy updates let the policy chase the **dense shaping** reward (hack the format) while **true correctness decays**

Prediction to test: raising the group size K shrinks $\mathrm{Var}(\hat{g}) \propto 1/K \Rightarrow$ smoother updates and **milder over-optimisation**.

I.3 The smoking gun: reward decomposition



In **R0**, the correctness reward (`check_answer`) **decays** after ≈ step 1000 while **format-shaping stays pinned high** — textbook reward-hacking.

R1 (β=0.3) collapses entirely ≈ step 1700.

The stable K=8 runs (**R3b**, **R5**) keep correctness *high throughout*.

I.3 Two hypotheses, one controlled 2×2

(1) Variance. Raise $K \Rightarrow \mathrm{Var}(\hat{g}) \propto 1/K \Rightarrow$ less over-optimisation.

(2) Regularisation. The KL penalty pulls toward π_{ref}; its penalised optimum is

$$\pi^\star(a) \;\propto\; \pi_{\mathrm{ref}}(a)\, e^{\hat{A}(a)/\beta} \xrightarrow[\beta\to\infty]{} \pi_{\mathrm{ref}},$$

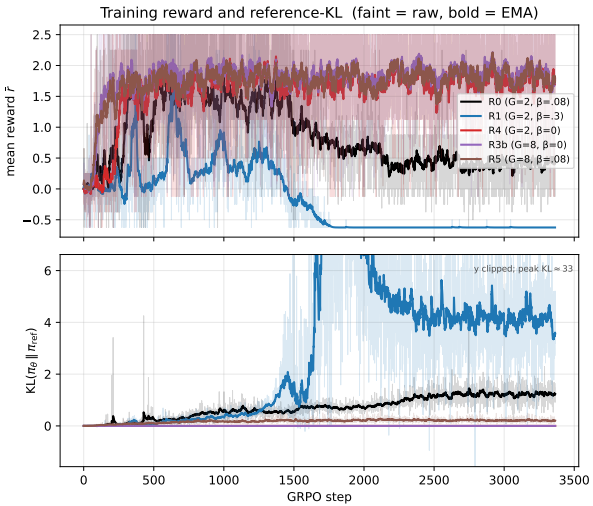
so a *tighter* leash (larger β) should curb the drift.

Design: $\beta \times K$ grid + a β-axis extension

	β=0	β=0.08	β=0.3
K=2	R4	**R0** (base)	R1
K=8	**R3b**	R5	—

5 full runs, all controlled: same data (HF GSM8K, seed 42), eval, horizon (3364 steps) and rollout RNG (`PRNGKey(0)`). *Only the named knob changes.*

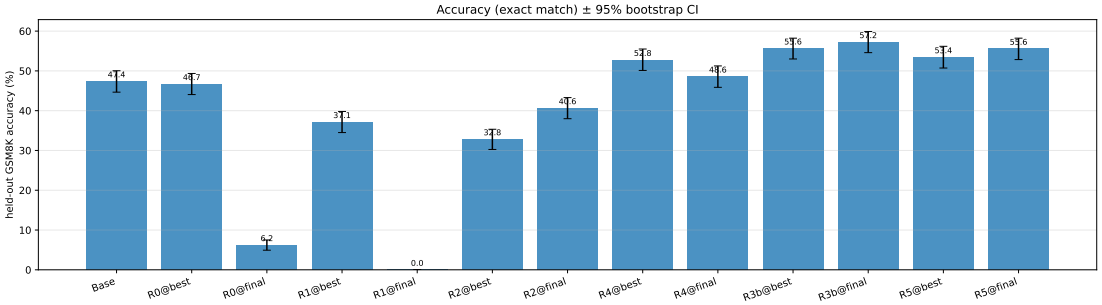
I.3 Training curves: reward and KL (all 5 runs)



- R0 over-optimises: reward peaks, then declines
- R1 (β=0.3) **collapses** — KL blow-up $\approx$ step 1700
- K=8 (R3b, R5) and R4 (β=0) stay **stable**

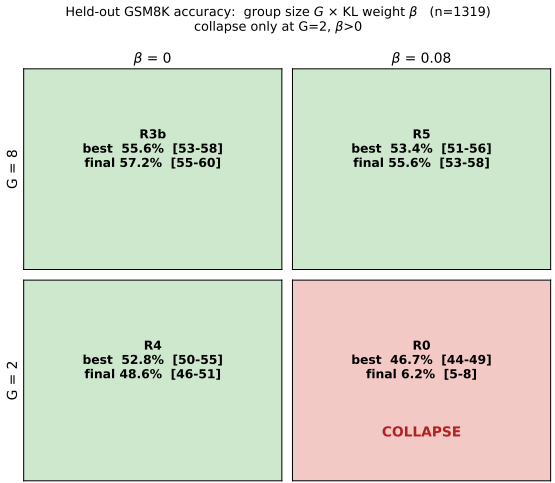
Note: a *higher* β did *not* buy a lower KL or more stability — the opposite of prediction (2).

I.3 Held-out accuracy (GSM8K test, n=1319)



- **R3b** (K=8, β=0): **57.2%** final, +9.9 pp over base — best model; paired 95% CI excludes 0
- Both K=8 runs beat base; each beats its matched K=2 run
- R0 final → 6.2%; R1 final → 0.0% (below base — a clean negative result)

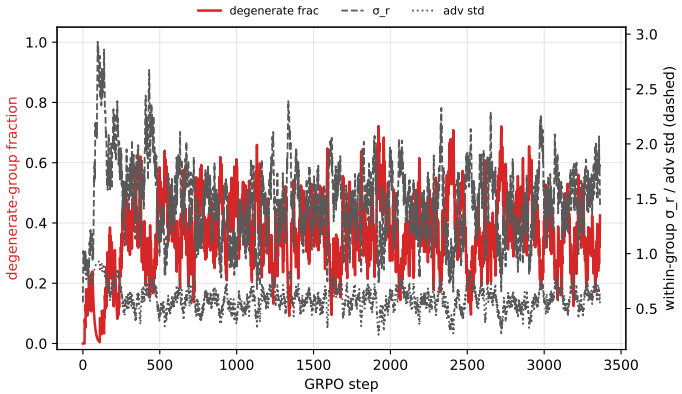
I.3 Collapse lives in one corner: K=2, $\beta > 0$



- Collapse occurs only at K=2, $\beta > 0$ (R0)
- K=2 **peaks then degrades** (R4 52.8 → 48.6; R0 collapses)
- K=8 **improves monotonically** to the final step

$\Rightarrow$ Over-optimisation is a K=2 **phenomenon** — exactly what the lower $\mathrm{Var}(\hat{g})$ predicts.

I.3 Diagnostic — the cost of K=8: group degeneracy



At K=8, about **a third of groups are degenerate** — all K rewards equal ⇒ **zero advantage**, wasted samples (the known dead-group / effective-sample-size failure mode).

Yet within-group σ_r **holds** ≈ 1.5 (steady, *not* $\to 0$).

The net effect is still +8–10 pp.

I.3 Findings vs. the theory

Prediction (1) holds

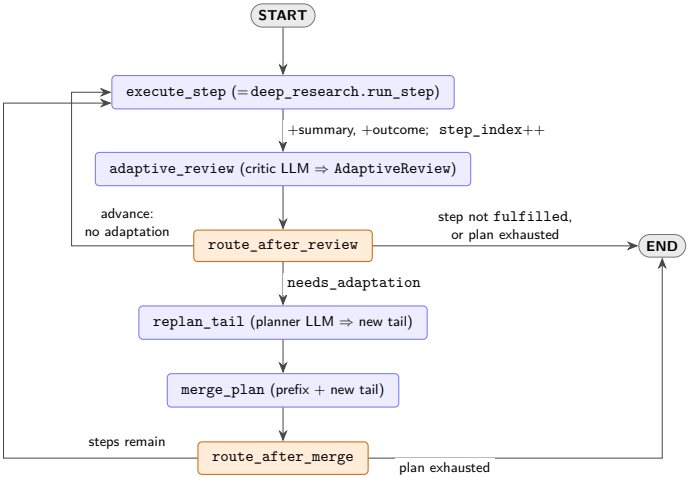
K=8 is the winning intervention. **Variance reduction** (group size) drives the +8–10 pp; over-optimisation is confined to K=2.

Prediction (2) contradicted

The tighter KL leash did *not* help: β=0 beats β=0.08 at K=2; β=0.3 collapses *faster*. At K=8 the KL weight is statistically irrelevant (R3b vs. R5: +1.7 pp, n.s.).

Punchline: the cure was **variance reduction** (larger groups), *not* regularisation — echoing DAPO's removal of the KL term for long-output RL.

Part II Adaptive planning in `cmbagent_lg`



Walkthrough.

- **Triggers a revision:** `adaptive_review` (critic LLM) sets `needs_adaptation`; the router replans *iff* the step was `fulfilled`, the plan is *not* exhausted, *and* adaptation is needed.
- **What may change:** *only* the **tail** (unrun steps); `merge_plan` = prefix + new tail.
- **Held fixed:** the completed prefix, its summaries / outcomes, and `step_index`.
- **Terminates:** when a step is not `fulfilled` or the plan is exhausted.

Part II Problems → proposed fixes

Problem	Fix
P1 Termination not *guaranteed* — the replanned tail can grow unboundedly; no monotone-decreasing quantity; `recursion_limit` *raises*, not finishes.	→ Thread a **revision budget / max-steps**; routers finish gracefully; reject over-long tails. *(A hard cap can truncate a genuine expansion.)*
P2 Adaptation only on *success* — a `fulfilled=False` step routes straight to END; no retry or route-around.	→ Add a **recovery / repair path** on failure with a per-step retry budget. *(Needs a recoverable-vs-fatal flag.)*
P3 Replanned tail *bypasses* the reviewer that gates the initial plan — `merge_plan` splices it unchecked.	→ Route `new_tail` through the **existing `plan_reviewer`** for one critique round. *(One extra reviewer call.)*
P4 Partial-state feedback at unconditional cost — sees only the *last* step (Markov); fires every step.	→ **Full-history, gated** review — cheap model / skip when no downstream impact. *(Gate conservatively.)*